Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.20	20.07	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.02	2.01	-0.7%
power (W), mean	629	642	+2.1%
power (W), peak	650	682	+5.1%
energy (Wh)	212	215	+1.4%
energy, parked workers still on at idle power (Wh)	212	215	+1.4%
CPU used (cores), mean	0.876	1.24	+41.1%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.037	0.052	+41.1%
energy per core-hour (Wh)	717	519	-27.6%
node-hours per core-hour	6.85	4.85	-29.1%
pending pods, pod-minutes	0.267	0.533	+100.0%
pending pods, peak	1.00	2.00	+100.0%
HPA replicas, mean	8.91	9.43	+5.8%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.800	0.517	-35.4%
requests timed	1,035	1,100	+6.3%
time over the response line (% of samples)	1.16	0.091	-92.2%
response time (ms), mean	164	95.44	-41.8%
response time (ms), median	132	84.60	-36.2%
response time (ms), 95th percentile	336	153	-54.7%
response time (ms), 99th percentile	505	171	-66.1%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 9; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.4769	5.3534	-28.4%	[-3.8263, -0.3857]	GAIN
kwh_per_core_hour	0.7798	0.5687	-27.1%	[-0.3784, -0.0421]	GAIN
utilisation	0.0360	0.0525	+45.8%	[+0.0034, +0.0304]	GAIN
pending_pod_minutes_per_hour	0.7619	1.6931	+122.2%	[-2.2857, +5.0794]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	2.2918	1.6402	-28.4%	[-3.6466, +2.4974]	EVEN, no significant difference

What Omni-Compass did

• decisions: 80 (one per minute)

• HPA target writes: 0

• node-pool resizes: 3

• scheduling-floor adds (pending pods): 0

• power-cap pod resizes (in place): 0

• rollout actions: 4

• power cap decided per minute: 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 0.695 0.695 0.695

Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	615	620	5.00	6.00	0.000	0.000	0.538	0.650
1	6.00	6.00	619	621	5.25	7.00	0.000	0.000	0.611	0.694
2	6.00	6.00	620	625	6.00	8.00	0.000	0.000	0.603	0.778
3	6.00	6.00	621	632	6.00	8.00	0.250	0.500	0.691	0.988
4	6.00	6.00	632	653	9.00	10.00	0.000	0.000	0.976	1.51
5	6.00	6.00	632	647	9.00	10.00	0.000	0.000	0.964	1.36
6	6.00	6.00	632	650	9.00	10.00	0.000	0.000	0.965	1.44
7	6.00	6.00	646	673	9.50	10.00	0.000	0.000	1.34	2.06
8	6.00	6.00	644	670	10.00	10.00	0.000	0.000	1.31	2.04
9	6.00	6.00	645	673	10.00	10.00	0.000	0.000	1.31	2.05
10	6.00	6.00	648	669	10.00	10.00	0.000	0.000	1.40	1.94
11	6.00	6.00	621	623	10.00	10.00	0.000	0.000	0.667	0.745
12	6.00	6.00	618	625	10.00	10.00	0.000	0.000	0.580	0.768
13	6.00	6.00	623	630	10.00	10.00	0.000	0.000	0.724	0.913
14	6.00	6.00	635	649	10.00	10.00	0.000	0.000	1.06	1.46
15	6.00	6.00	634	648	10.00	10.00	0.000	0.000	1.05	1.41
16	6.00	6.00	632	645	10.00	10.00	0.000	0.000	0.941	1.33
17	6.00	6.00	626	636	10.00	10.00	0.000	0.000	0.819	1.08
18	6.00	6.00	618	626	10.00	10.00	0.000	0.000	0.599	0.791
19	6.00	6.00	616	624	10.00	10.00	0.000	0.000	0.560	0.751

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.